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Abstract

It is crucial for enterprises to clearly identify user needs during the process of formulating product design improvement plans. Therefore, it is essential to comprehensively and accurately identify user needs, explore the reasons behind the emergence of these needs, and incorporate user opinions into the process of product design improvement. A method is proposed to comprehensively and accurately capture user requirements and address the challenge of identifying the underlying causes of user requirements. This method utilizes online comments and operational data to identify user requirements and their influencing factors. First, text sentiment analysis techniques are employed to quantify the importance and performance values of product feature topics. Second, we construct a quadrant model to identify product features requiring improvement, and the original negative comments related to these features are traced. However, the quadrant model alone is insufficient to reflect specific product issues that users are concerned about. Therefore, a functional structure model based on product issues is designed to filter and identify factors that influence user requirements using operational data. Finally, a Bayesian network inference approach is utilized to identify the key influencing factors on user requirements, enabling analysis of the causes behind user requirements and the proposal of product design improvement strategies. The feasibility and effectiveness of the proposed method are validated through experiments conducted on heavy-duty truck data. By analyzing the original negative comments related to the power characteristics, specific user demands regarding the insufficient power of the product were identified, such as “obviously insufficient power when climbing slopes” and other issues. Based on the vehicle power system

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related to the state of parts and user behavior that may affect “insufficient vehicle power” were identified. Based on the analysis results, suggestions were made to improve the engine intake air temperature control strategy and to enhance vehicle performance by promoting correct user behavior through informational campaigns.

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Keywords: [Online review](#) ▪ [operation data](#) ▪ [user requirements](#) ▪ [Bayesian network](#)

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